

1(a)(i)	0.40 ;	1
1(a)(ii)	fucoxanthin ; diadinoxanthin ;	2
1(b)	<i>blue light:</i> 1 <i>S. costatum</i> (will have), greater / faster / AW , (rate of photosynthesis than <i>R. salina</i>) ; ora <i>green light:</i> 2 <i>S. costatum</i> (will have), slightly / AW, greater / faster / AW , (rate of photosynthesis than <i>R. salina</i>) ; ora <i>red light:</i> 3 <i>S. costatum</i> (will have), the same / similar / AW, (rate of photosynthesis as <i>R. salina</i>) ;	3
1(c)(i)	any one from: 1 <i>idea that</i> absorbance measured (with colorimeter) / colour of indicator solution, is not affected by algae ; 2 indicator solution can be separated from algal balls ; 3 (more accurate) standardisation of, mass / volume / AW, of algae ;	1
1(c)(ii)	<i>independent variables:</i> species / type, (of algal beads) ; wavelength / colour, of light ; <i>dependent variable:</i> absorbance / pH (of solution) ;	3

Question	Answer	Marks
1(c)(iii)	any eight from: 1 <u>method</u> for use of filters (to give blue, green, red coloured light) ; 2 use, same / stated, number / mass / volume, of algal beads (in each bottle) ; 3 place light source the, same / stated, distance (from algal beads) ; 4 carry out experiment in a darkened room ; 5 use, same / stated, volume / concentration, of indicator (in each bottle) ; 6 (use of a control bottle) replace the algal beads with, glass / plastic / AW, beads ; 7 use, red / pH 8.4, hydrogencarbonate indicator solution (at start) ; 8 detail of <u>method</u> to take sample (of indicator solution) from bottle ; 9 use a, same / stated, coloured, filter / light, in the colorimeter ; 10 <u>method</u> to calibrate colorimeter ; 11 <i>ref. to</i>, suitable time to leave algal beads and indicator solution (before measuring absorbance) ; 12 <u>for each colour (filter) and species</u>, measure / record / AW, absorbance, at same / stated, time(s) ; 13 use at least three measurements <u>for each colour (filter) and species</u>, and calculate a mean ;	8

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1(c)(iv)	hazard and risk and precaution ; <table border="1"> <thead> <tr> <th>hazard</th><th>risk</th><th>precaution</th></tr> </thead> <tbody> <tr> <td>algae / beads</td><td>irritant / allergy</td><td>gloves / mask / goggles / PPE</td></tr> <tr> <td>hydrogencarbonate indicator (solution)</td><td>irritant / allergy</td><td>gloves / mask / goggles / PPE</td></tr> <tr> <td>heat from light source</td><td>burns</td><td>do not touch bulb / turn off lamp before handling</td></tr> </tbody> </table>	hazard	risk	precaution	algae / beads	irritant / allergy	gloves / mask / goggles / PPE	hydrogencarbonate indicator (solution)	irritant / allergy	gloves / mask / goggles / PPE	heat from light source	burns	do not touch bulb / turn off lamp before handling	1
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1(d)(i)	any one from: 1 comparing the means of two (sets of data) ; 2 continuous data (collected) ; 3 (data are from populations that are) normally distributed ; 4 standard deviations are approximately the same ;	1												
1(d)(ii)	there is no difference, in the pH of samples (taken from the small bottles / of indicator solution), between <i>R. salina</i> and <i>S. costatum</i> ;	1												